

9 CLEANUP PROCEDURES

9.1 Summary

This chapter describes cleaning procedures for each component.
Clean the components properly, depending on weighed product types and dirt conditions.
Before cleaning, drain any weighed products remaining in the device (Refer to "6.7 Drain screen").

**WARNING**

- **When cleaning each component, the operator must turn OFF and lock the main power switch, and keep the key in his possession during the work.**
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<Contents>

- Cleaning procedures and cycle of the components

<Purpose>

- To understand installation, removal, and cleaning procedures of the components in order to prevent foreign matter from entering the product and to maintain the hygiene of the device.

<Intended reader>

- Operators
- Maintenance engineers

9.2 Washing and Sterilization

Although the weigher has a water proof structure (IP-66), physical damage may occur if the weigher is washed outside the specified range or is incorrectly handled.

The maintenance supervisor should understand the washing methods, and check that the components are cleaned by the operators in accordance with the washing methods.



WARNING

- **Do not wash the components using any methods other than as instructed below.**

Washing and drying methods

This device consists of removable units and unremovable units.

For the cleaning of the removable units, always remove from the device, and wash and dry by following the relevant washing method described in "9.4.1 Washing Removable Unit".

For the unremovable unit, wash and dry by following the relevant washing method described in "9.4.2 Cleaning the Unremovable Unit" and in accordance with the items below.

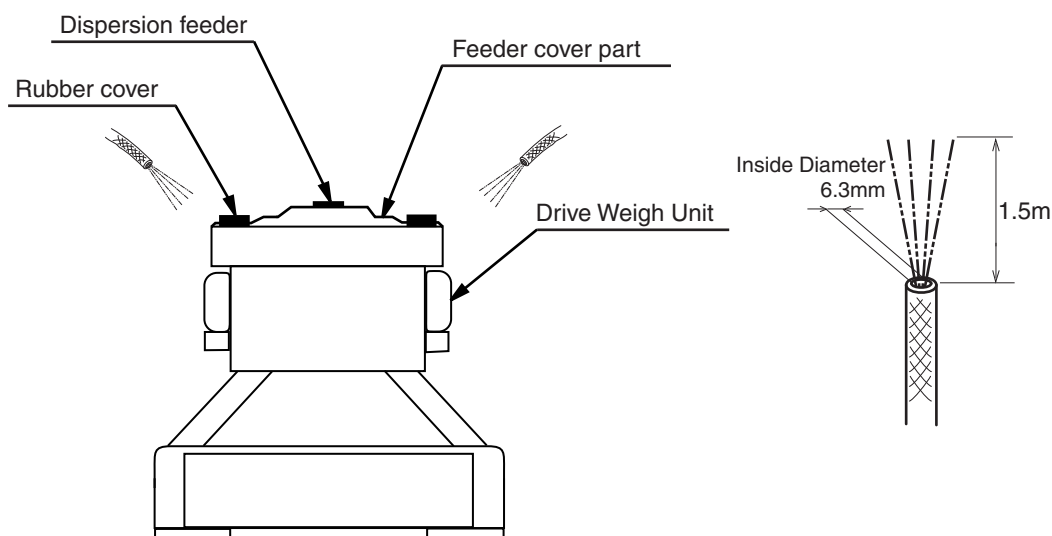


Fig. 9-1

1. Use a soft cloth to remove the residue from each component surface.
 2. Washing method
 - a. Wipe the surface first with a soft cloth soaked with the specified detergent (neutral detergent), and then with a damp cloth with water.
 - b. When using a hose, spray the specified detergent (neutral detergent), rinse with the following amount of water and wipe with a soft cloth.
- Use the hose with 6.3 mm in inner diameter. Set the water so that it rises up to 1.5 m when the hose end is directed upward
(Water volume that collects approximately 10.5 l/min at water pressure of approximately 18 kPa).

CAUTION

- Use a neutral detergent for washing. Choose a detergent that does not affect silicon materials.

3. To dry after washing, wipe the surface with a soft cloth.

4. To prevent product residue from adhering, wash and dry the components daily.

CAUTION

- Avoid overload to the weigh drive unit and dispersion feeder. These components are interlocked to the weight sensor.
- For cleaning of the rubber cover installed on the feeder cover, do not use tools such as a brush or any tools with sharp parts.

Washing and drying methods in the vicinity of the weigh drive unit

The previous section described the washing and drying methods for the unremovable units. As for the weigh drive unit, especially because precision parts are installed around the unit, wash and dry following the precautions below, in addition to the washing and drying methods for components.

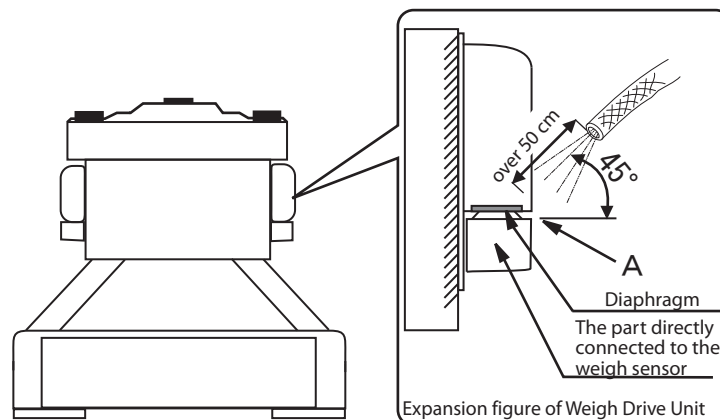


Fig. 9-2

Maintain a hose angle of approximately 45° during washing, and keep a distance of 50 cm or more from the components.

NOTE

- At position A in the above drawing, a diaphragm is connected to the weight sensor. This part is extremely sensitive. When any washing method other than instructed is used, malfunctions may occur due to liquid intrusion.
1. Do not use air apparatus with nozzles to dry out position A. It may break the diaphragm.
 2. Do not use tools such as a brush or any tools with sharp parts to dry out position A. It may break the diaphragm.

9.3 Washing and Sterilization Methods

The recommended standard methods for washing and sterilizing are described in "Table 9-1". For the details of each component, read "9.4.2 Cleaning the Unremovable Unit" thoroughly.

CAUTION

- **Washing and sterilizing methods differ depending on product types, processing methods, and bacterial conditions.**
Create a washing and sterilization manual, and perform washing and sterilization work daily. For creating a manual, refer to "Table 9-1". For details, contact the distributor or Ishida customer support.
- **For the unremovable units and parts, washing and sterilization methods differ depending on the machine specifications (mainly the specifications of waterproof and dustproof construction).**
This can be verified by determining whether the name of your device ends with "WP", "PB" or "SS".
For details, contact the distributor or Ishida customer support.
[Example]
CCW-RV-*-WP:equivalent to IP66
CCW-RV-*-PB, SS:equivalent to IP50
- **After washing and sterilization, always dry. Failure to do so may cause microorganism development.**
- **If spot welding causes the development of bacteria, contact the distributor or Ishida customer support.**

Table 9-1

Unit and Parts	Washing Method	Sterilization Method
<Removable unit> • Inlet chute • Dispersion table • Radial trough • Pool hopper • Weigh hopper • Collection chute • Discharge chute • Timing hopper	<Immersion cleaning> Soak the unit and parts in the specified detergent (neutral detergent). After removing, brush and rinse with water. NOTE • Use a soft brush.	<Immersion sterilization> Soak the unit and parts in warm water or disinfectant for sterilization, then rinse with water and dry. 1. Warm water Soak approximately 30 minutes in 80 to 90 deg C water. 2. Disinfectant Soak approximately 20 minutes in sodium hypochlorite (available chlorine concentration 250 ppm, 20 to 25 deg C).

Table 9-1

Unit and Parts		Washing Method	Sterilization Method
<Unremovable unit> • Dispersion feeder • Dispersion cover • Weigh drive unit • Main body, parts other than above • Remote control unit	WP specification (IP66)	<Brushing wash or wipe wash> 1.Brushing wash Using a brush with the specified detergent (neutral detergent), brush the surface, and then wipe with a damp cloth in water. 2. Wiping wash Wipe with a cloth with the specified detergent (neutral detergent), and then wipe with a damp cloth in water. NOTE • Use a soft brush. • For the remote control unit, wipe wash only. • Do not use a brush to the rubber cover of the radial feeder. • Avoid overload to the dispersion feeder and the weigh hopper hanger portion of the weigh drive unit. These components are interlocked to the weight sensor.	<Spray sterilization or wipe sterilization> 1.Spray sterilization Spray the disinfectant by a sprayer. 2. Wiping sterilization Wipe with a cloth with disinfectant. * Disinfectant • Sodium hypochlorite (available chlorine concentration 250 ppm) • Alcohol:80$\frac{v}{v}$% NOTE • Avoid overload to the dispersion feeder and the weigh hopper hanger portion of the weigh drive unit. These components are interlocked to the weight sensor. • For the remote control unit, wipe and sterilize with alcohol:80$\frac{v}{v}$% only. • After sterilizing, dry.

Table 9-1

Unit and Parts		Washing Method	Sterilization Method
<Unremovable unit> • Dispersion feeder • Dispersion cover • Weigh drive unit • Main body, parts other than above • Remote control unit	PB, SS specification (IP50)	<Wipe wash> 1. Wiping wash Wipe with a cloth with the specified detergent (neutral detergent), and then wipe with a damp cloth in water.	<Wipe sterilization> 1. Wiping sterilization Wipe with a cloth with disinfectant. * Disinfectant • Alcohol:80 $\frac{v}{v}$ %
		NOTE • Avoid overload to the dispersion feeder and the weigh hopper hanger portion of the weigh drive unit. These components are interlocked to the weight sensor.	NOTE • Avoid overload to the dispersion feeder and the weigh hopper hanger portion of the weigh drive unit. These components are interlocked to the weight sensor. • After sterilizing, dry.

9.4 Washing and Sterilizing of Components

This section describes how to wash and sterilize components.

9.4.1 Washing Removable Unit

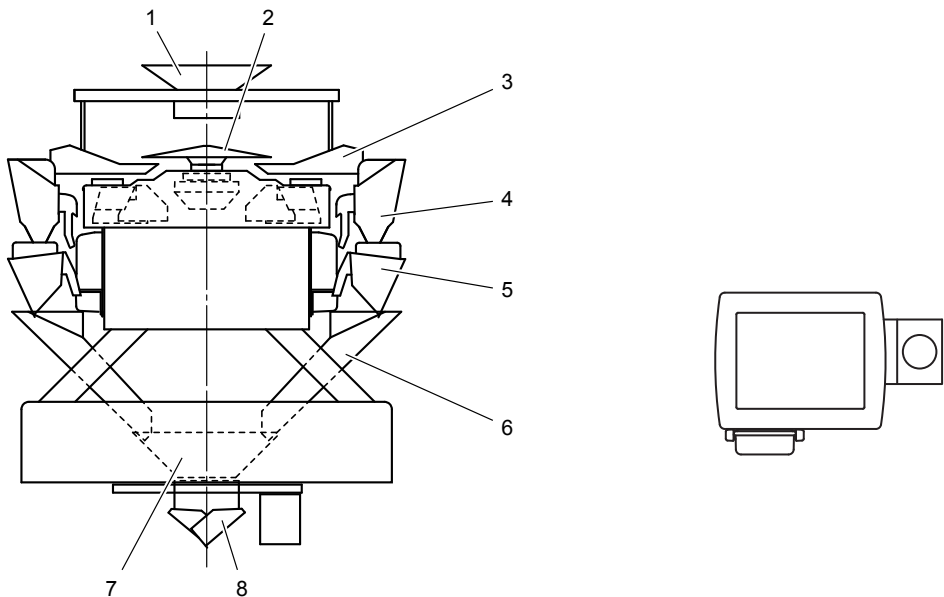


Fig. 9-3

Table 9-2

No.	Name
1	Inlet chute
2	Dispersion table
3	Radial trough
4	Pool hopper
5	Weigh hopper
6	Collection chute
7	Discharge chute
8	Timing hopper

9.4.1.1 Inlet Chute

1. Loosen the butterfly nut to remove the inlet chute.
2. After washing and sterilizing, dry (For the washing and sterilization methods, refer to "Table 9-3 ").
3. Reinstall in the reverse order of the removal.
4. Make sure the inlet chute is installed securely.

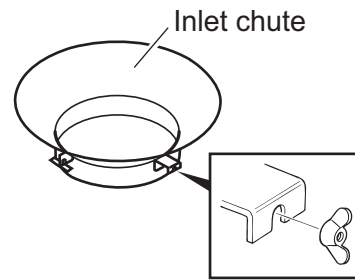


Fig. 9-4

Table 9-3

Washing method	Immersion cleaning (Neutral detergent).
Sterilization method	Immersion sterilization. Warm water (80 to 90°C): 30 minutes. Or, sodium hypochlorite (available chlorine concentration 250 ppm): 20 minutes.

9.4.1.2 Dispersion Table



CAUTION

- When cleaning the dispersion table, do not push it down. Doing so may damage the weight sensor.

1. Turn the dispersion table counterclockwise (turn to the left).
2. After washing and sterilizing, rinse with water and dry (For the washing and sterilization methods, refer to "Table 9-4 ").
3. Reinstall in the reverse order of the removal.
4. Make sure the dispersion table is installed securely.

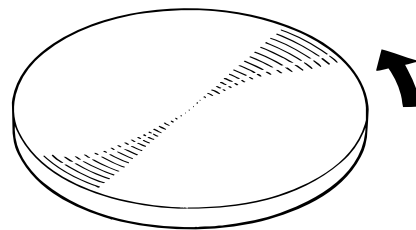


Fig. 9-5

NOTE

- Perform the installation of the dispersion table after installing the radial troughs

Table 9-4

Washing method	Immersion cleaning (Neutral detergent).
Sterilization method	Immersion sterilization. Warm water (80 to 90°C): 30 minutes. Or, sodium hypochlorite (available chlorine concentration 250 ppm): 20 minutes.

9.4.1.3 Radial Trough

NOTE

- Remove the radial troughs after removing the pool hoppers.

1. Pull up the clamp lever at the bottom of the radial trough, and remove the radial trough.
2. After washing and sterilizing, rinse with water and dry (For the washing and sterilization methods, refer to "Table 9-5").

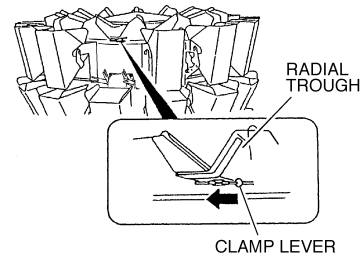


Fig. 9-6

3. Reinstall in the reverse order of the removal.
4. Make sure the radial troughs are not touching each other.
5. If there is any contact, repeat installation so that they do not touch each other.

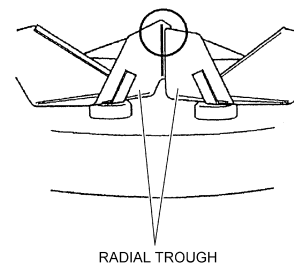


Fig. 9-7

Table 9-5

Washing method	Immersion cleaning (Neutral detergent).
Sterilization method	Immersion sterilization Warm water (80 to 90°C): 30 minutes. Or, sodium hypochlorite (available chlorine concentration 250 ppm): 20 minutes.

9.4.1.4 Pool Hopper and Weigh Hopper

CAUTION

- When installing and removing the weigh hopper, do not apply an excessive load.
Doing so may damage the weight sensor.

NOTE

- When installing and removing the hoppers, securely support the side of the hoppers with both hands.

1. Pull the pool hopper forward to remove.

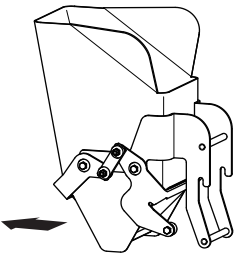


Fig. 9-8

2. Pull the weigh hopper forward to remove.

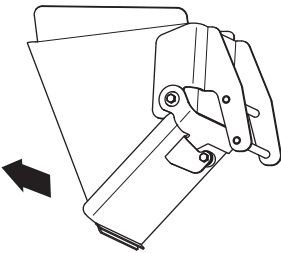


Fig. 9-9

3. After washing and sterilizing, rinse with water and dry (For the washing and sterilization methods, refer to "Table 9-6 ").

NOTE

- When the radial troughs are removed, install them first.

4. Reinstall in the reverse order of the removal.
5. Make sure the roller of each hopper fits into the switching lever.
6. If not, install it again.

Table 9-6

Washing method	Immersion cleaning (Neutral detergent).
Sterilization method	Immersion sterilization. Warm water (80 to 90°C): 30 minutes. Or, sodium hypochlorite (available chlorine concentration 250 ppm): 20 minutes.

9.4.1.5 Collection Chute

NOTE

- Install the collection chute after installing the discharge chute.

1. Pull up the collection chute upper portion, and lift up to remove.
2. After washing and sterilizing, rinse with water and dry (For the washing and sterilization methods, refer to "Table 9-7").
3. Reinstall in the reverse order of the removal.
4. Make sure that the collection chute is securely installed.

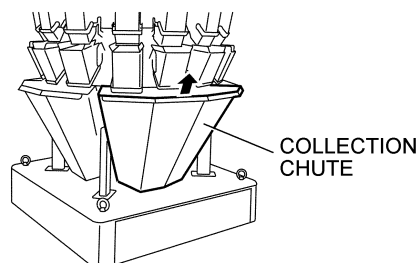


Fig. 9-10

Table 9-7

Washing method	Immersion cleaning (Neutral detergent).
Sterilization method	Immersion sterilization. Warm water (80 to 90°C): 30 minutes. Or, sodium hypochlorite (available chlorine concentration 250 ppm): 20 minutes.

9.4.1.6 Discharge Chute

NOTE

- Remove the discharge chute after removing the collection chute.
- Removal procedures of the discharge chute differ depending on the devices.

1. Lift up the discharge chute, and remove between the legs.
For the type that the discharge chute cannot be removed between the legs, remove the timing hopper first, and then remove the discharge chute from underneath (Refer to "9.4.1.7 Timing Hopper").

2. After washing and sterilizing, rinse with water and dry (For the washing and sterilization methods, refer to "Table 9-8").
3. Reinstall in the reverse order of the removal.
4. Make sure the discharge chute is installed securely.

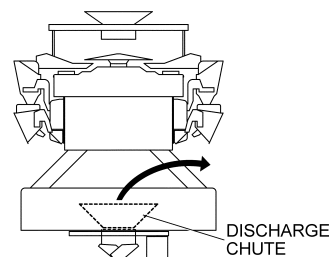


Fig. 9-11

Table 9-8

Washing method	Immersion cleaning (Neutral detergent).
Sterilization method	Immersion sterilization. Warm water (80 to 90°C): 30 minutes. Or, sodium hypochlorite (available chlorine concentration 250 ppm): 20 minutes.

9.4.1.7 Timing Hopper

1. Loosen the butterfly nut of the drive shaft, and disconnect the timing hopper from the drive unit.

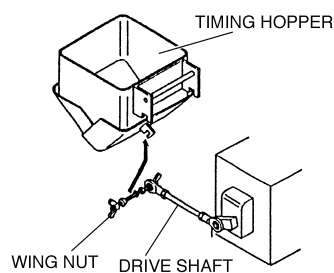
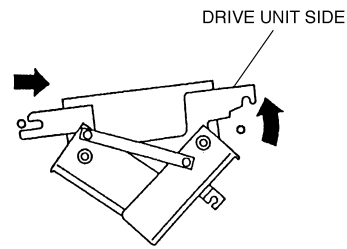


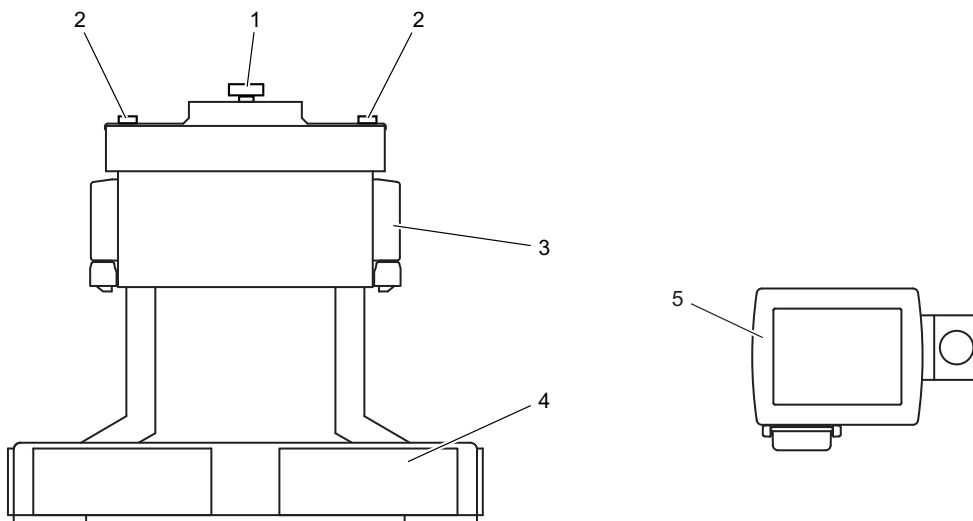
Fig. 9-12

2. Slightly lift the drive unit side of the timing hopper, and remove the timing hopper.
3. After washing and sterilizing, rinse with water and dry (For the washing and sterilization methods, refer to "Table 9-9").
4. Reinstall in the reverse order of the removal.
5. Make sure the timing hopper is installed securely.

**Fig. 9-13****Table 9-9**

Washing method	Immersion cleaning (Neutral detergent).
Sterilization method	Immersion sterilization. Warm water (80 to 90°C): 30 minutes. Or, sodium hypochlorite (available chlorine concentration 250 ppm): 20 minutes.

9.4.2 Cleaning the Unremovable Unit

**Fig. 9-14****Table 9-10**

No.	Name
1	Rubber cover of the dispersion feeder (Refer to "9.4.2.1 Main Body, Dispersion Portion, and Drive Unit")
2	Rubber cover of the radial feeder (Refer to "9.4.2.1 Main Body, Dispersion Portion, and Drive Unit")
3	Drive unit (Refer to "9.4.2.1 Main Body, Dispersion Portion, and Drive Unit")
4	Main body (Refer to "9.4.2.1 Main Body, Dispersion Portion, and Drive Unit")
5	Remote control unit (Refer to "9.4.2.2 Remote Control Unit")

9.4.2.1 Main Body, Dispersion Portion, and Drive Unit

CAUTION

- The precautions on washing of the main body and drive portion are shown in "9.2 Washing and Sterilization". Wash and dry following the instructions.

NOTE

- Do not use a brush to wash the rubber covers of the dispersion feeder and radial feeder.
- Avoid overload to the dispersion feeder and WH hanger. These components are interlocked to the weight sensor.

1. Remove the parts below by following the procedures in "9.4.1.1" to "9.4.1.7".

- Inlet chute
- Dispersion table
- Radial trough
- Pool hopper
- Weigh hopper
- Collection chute
- Discharge chute
- Timing hopper

2. In accordance with the specifications, wash, sterilize and dry the device (For the washing and sterilization methods, refer to "Table 9-11 WP Specification" and "Table 9-12 PB, SS Specification").
3. Reinstall in the reverse order of the removal.
4. Make sure that each part is installed securely.

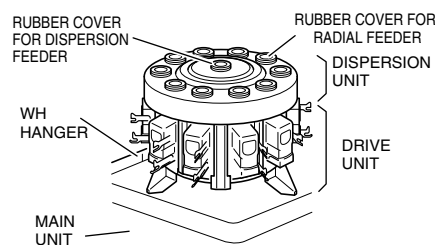


Fig. 9-15

Table 9-11 WP Specification

Washing method	Brushing wash or wipe wash (Neutral detergent).
Sterilization method	Spray sterilization or wipe sterilization. Alcohol:80 $\frac{v}{v}$ %. Or, sodium hypochlorite (available chlorine concentration 250 ppm).

Table 9-12 PB, SS Specification

Washing method	Wipe wash (Neutral detergent).
Sterilization method	Wipe sterilization. Alcohol:80 $\frac{v}{v}$ %.

9.4.2.2 Remote Control Unit

 CAUTION

- Do not rub hard or use adhesive tape to remove dirt when cleaning the operation screen.
- Do not clean the operation screen with organic solvents such as thinner or benzin.
- Do not pour detergent directly over the operation screen.

1. After washing and sterilizing, wipe with a damp cloth in water and dry (For the washing and sterilization methods, refer to "Table 9-13 ").

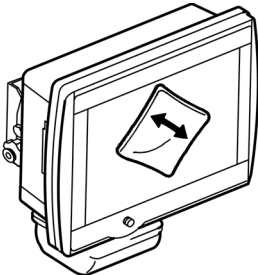


Fig. 9-16

Table 9-13

Washing method	Wipe wash (Neutral detergent).
Sterilization method	Wipe sterilization. Alcohol:80 ^v / _v %.

